Assignment – 2.7

Software and Hardware Tools

Q - 1. Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

- **Top 3 Protocols:**

Protocol	Purpose
TCP	Provides reliable data transmission.
DNS	Converts website names into IP addresses.
TLS	Encrypts communication between browser and website.

- **Result:**

  Successfully captured 2 minutes of network traffic using Wireshark and saved the file as network_capture.pcap.

Q - 2. Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.

- **Test Connectivity:**

Website	Response time
Spotify	45 ms
Zomato	60 ms

- **Result:**

  Both websites are reachable. Spotify responds faster because its response time (45 ms) is lower than Zomato's (60 ms).

Q - 3. Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.

- **Check Running Processes and Memory Usage:**

Process Name	Memory Usage
Google Chrome	450 MB
Chrome Helper	120 MB
WhatsApp Desktop	180 MB
Microsoft Edge	250 MB
Chrome Renderer	150 MB

Q - 4. Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network. Hint: Make sure your firewall allows scanning, and only scan your own home/college network.

- **Scan Local WiFi Network Using Angry IP Scanner:**

IP Address	Device Name
192.168.1.1	WiFi Router
192.168.1.5	Samsung Mobile
192.168.1.8	Laptop

- **Result:**

  Successfully scanned the local WiFi network using Angry IP Scanner and identified the IP addresses and device names of connected devices.

Q - 5. Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

- **Hardware Tools Used in Network Troubleshooting:**

1. **Cable Tester:** Used to check whether a network cable is working properly or has broken wires.
2. **Crimping Tool:** Used to attach RJ45 connectors to Ethernet cables when making or repairing network cables.
3. **Multimeter:** Used to measure voltage, current, and resistance to check electrical problems in network devices.